ACADEMIC CASE STUDY SPECIFICATION

Sports Tournament Ranking System

1. PROBLEM STATEMENT

Develop a Python program to store match results, calculate team points, apply tie-break rules, and display tournament rankings.

2. OBJECTIVES

- Store match results.
- Calculate team points.
- Calculate goal difference.
- Apply tie-break rules.
- Display team rankings.

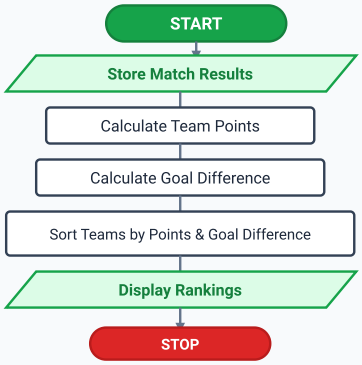
3. ASSUMPTIONS

- Win = 3 points.
- Draw = 1 point.
- Loss = 0 points.
- Goal difference is used as the first tie-breaker.
- Higher points and goal difference get higher rankings.

4. ALGORITHM

1. Start.
2. Store match results.
3. Initialize team points and goal difference.
4. Calculate points for each match.
5. Update goal difference.
6. Sort teams by points and goal difference.
7. Display rankings.
8. Stop.

5. FLOWCHART



6. PSEUDOCODE

```
START Store match results FOR each match: IF team 1 wins: Add 3 points to team 1 ELSE IF match is draw: Add 1 point to both teams ELSE: Add 3 points to team 2 Calculate goal difference Sort teams by: Points descending Goal difference descending Display rankings STOP
```

7. PYTHON PROGRAM

```
matches = [
    ("India", "Australia", 3, 1),
    ("India", "England", 2, 2),
    ("Australia", "England", 2, 0)
]

teams = {}

for t1, t2, s1, s2 in matches:
    for team in [t1, t2]:
        if team not in teams:
            teams[team] = [0, 0]

    teams[t1][1] += s1 - s2
    teams[t2][1] += s2 - s1

    if s1 > s2:
        teams[t1][0] += 3
    elif s1 < s2:
        teams[t2][0] += 3
    else:
        teams[t1][0] += 1
        teams[t2][0] += 1

ranking = sorted(
    teams.items(),
    key=lambda x: (x[1][0], x[1][1]),
    reverse=True
)

print("TOURNAMENT RANKINGS")
for i, (team, data) in enumerate(ranking, 1):
    print(i, team, "Points:", data[0], "Goal Difference:", data[1])
```

8. SAMPLE OUTPUT

```
TOURNAMENT RANKINGS

1 India Points: 4 Goal Difference: 2
2 Australia Points: 3 Goal Difference: 0
3 England Points: 1 Goal Difference: -2
```

9. TEST CASES

Test #	Match Result	Expected Result
1	3–1	Winner gets 3 points
2	2–2	Both get 1 point
3	0–2	Second team gets 3 points
4	Equal points	Compare goal difference
5	Different points	Higher points ranked first

10. CONCLUSION

The Sports Tournament Ranking System calculates team points and goal differences, applies tie-break rules, and displays rankings using Python lists, dictionaries, loops, and sorting.

Sample Industry Use Cases & Application Examples:

- **League Standings Automation:** Automates real-time point tables and rank calculations for sports tournaments like football, cricket, and esports leagues.
- **Data Validation & Ranking Logic:** Demonstrates multi-criteria sorting (points followed by goal difference) to handle complex tie-breaker scenarios effectively.
- **Scalable Data Management:** Uses Python data structures (dictionaries & lists) for dynamic team additions and seamless score tracking.